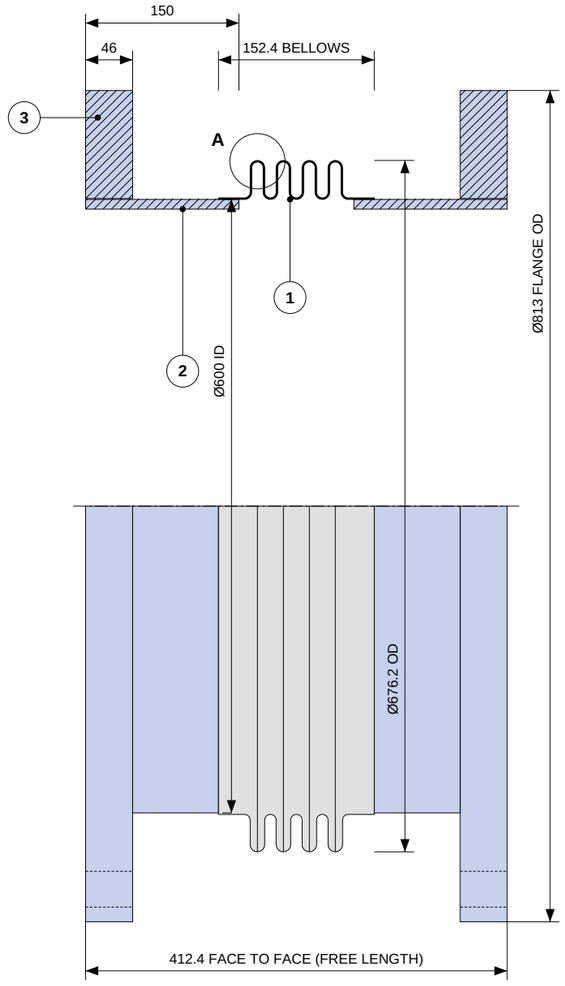
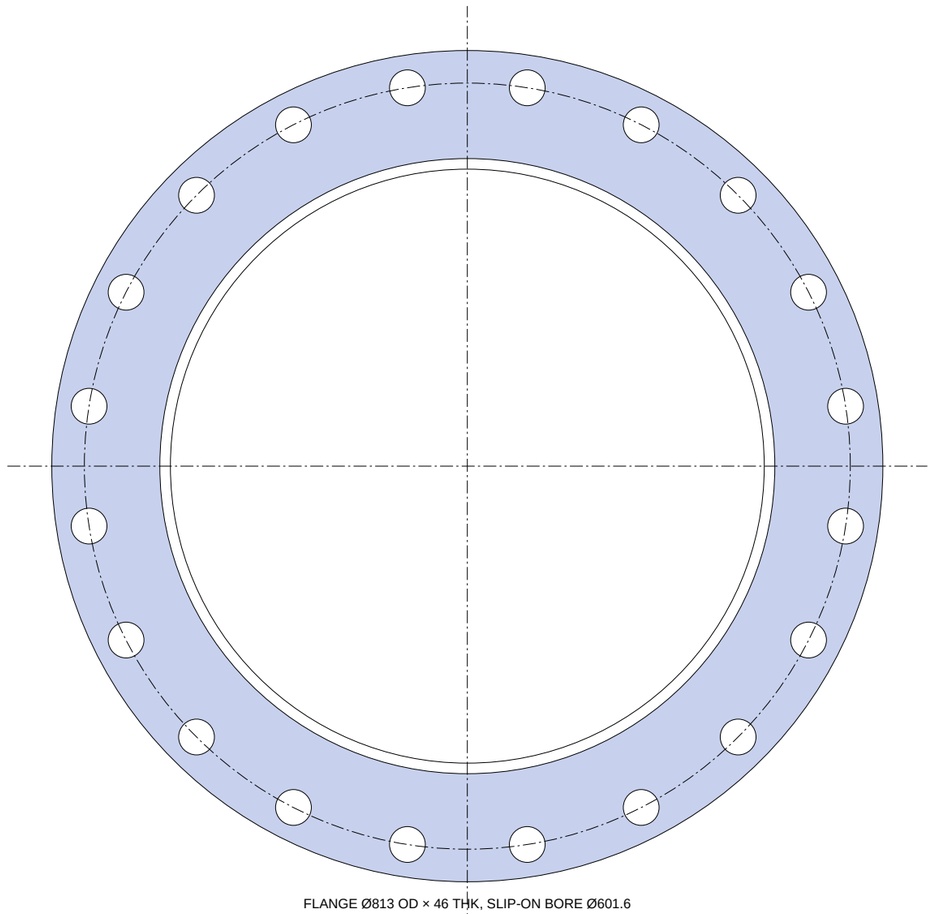
ELEVATION — UPPER HALF IN SECTION (1:6)

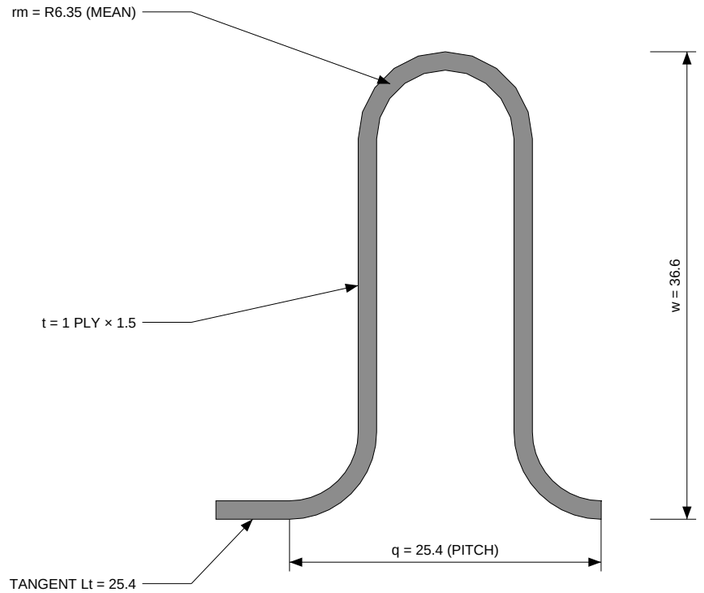


VIEW ON FLANGE FACE (1:6)



FLANGE Ø813 OD × 46 THK, SLIP-ON BORE Ø601.6
20 HOLES Ø35 EQ. SP. ON Ø749.3 PCD, STRADDLING CENTRELINES
DRILLING TO ASME B16.5 CLASS 150 NPS 24 · BOTH ENDS IDENTICAL

DETAIL A — CONVOLUTION (2:1)



U-36 FORMING DIE · N = 4 CONVOLUTIONS
Dm = 638.1 mm · tp = t·√(Db/Dm) = 1.455 mm

EJMA DESIGN DATA

Design code	EJMA Standards, 9th Ed. (2008) — bellows §4.13.1
Calculation ref.	EJ-2026-6BMPAW (design mode), 2026-10-12
Design pressure / temperature	5.0 bar(g) internal / 150 °C
Movement / cycles	25 mm axial compression / 1,000 cycles required
Bellows	Unreinforced U-profile, 1 ply × 1.5 mm, N = 4, SA-240 321
Allowable stress Sab (150 °C)	114.0 MPa (ASME II-D 2019, Table 1A)
Effective area Ae / pressure thrust	0.3198 m² / 159.9 kN (main anchors)
Spring rate (working, overall)	1394 N/mm · axial force at 25 mm: 34.9 kN
Equivalent movement e (per conv.)	6.25 mm

EJMA CHECKS — RESULT: ALL PASS

Check	Eq.	Value	Limit	Unit	Status
Tangent hoop S1	4-27	56.6	≤ 114.0	MPa	PASS
Bellows hoop S2	4-29	31.8	≤ 114.0	MPa	PASS
Meridional S3+S4	4-30/4-31	125.7	≤ 171.0	MPa	PASS
Fatigue life Nc	4-34	3015	≥ 1000	cycles	PASS
Column squirm Psc	4-35	5.00	≤ 218.79	bar(g)	PASS
In-plane squirm Psi	4-36	5.00	≤ 6.68	bar(g)	PASS
Max. compression ec	§4.2	6.25	≤ 11.20	mm	PASS
Axial instability ex	§4.7	6.25	≤ 1252.37	mm	PASS

GENERAL NOTES

1. DESIGN: BELLOWS DESIGNED TO EJMA STANDARDS 9TH ED. (2008) §4.12–4.13; MOVEMENTS PER §4.1–4.3. DIMENSIONS IN mm.
2. BELLOWS: SA-240 TYPE 321, 1 PLY × 1.5 mm, ANNEALED. LONGITUDINAL SEAM BUTT-WELDED, FULL PENETRATION, 100% PT (EJMA §7.1.2).
3. ATTACHMENT WELDS: BELLOWS TANGENT TO PIPE NIPPLE, CIRCUMFERENTIAL FILLET WELD ALL ROUND; PT AFTER WELDING.
4. PIPE NIPPLES ASTM A106 GR B / A53, Ø600 × 9.53 WALL. FLANGES: SLIP-ON, CARBON STEEL A105, CLASS 150 NPS 24 BOLT PATTERN, BORE Ø601.6 FOR Ø600 PIPE; FILLET WELDED INSIDE AND OUTSIDE.
5. HYDROSTATIC TEST: 7.5 bar(g) (1.5 × DESIGN). TEST PRESSURE SHALL NOT EXCEED 10.0 bar(g) = 1.5 × Psi (EJMA §4.12.1.6). JOINT MUST BE RESTRAINED AGAINST PRESSURE THRUST DURING TEST (EJMA §7.2.1).
6. PRESSURE THRUST 160 kN AND SPRING FORCE MUST BE RESISTED BY MAIN ANCHORS. PIPE GUIDES: 1ST ≤ 4D, 2ND ≤ 14D FROM THE JOINT (EJMA §2.10). DO NOT USE THE JOINT TO CORRECT PIPING MISALIGNMENT.
7. SHIPPING BARS (PAINTED YELLOW) TO BE FITTED FOR TRANSPORT AND REMOVED AFTER INSTALLATION, BEFORE PRESSURE TEST (EJMA §8.2). PROTECT BELLOWS FROM WELD SPATTER, ARC STRIKES AND MECHANICAL DAMAGE (EJMA §8.5).
8. FINISH: CARBON STEEL PARTS PRIMED + BLUE ENAMEL; BELLOWS NOT PAINTED. TOLERANCES PER EJMA §6.17.
9. PRELIMINARY — MATERIAL DATA ASME II-D 2019; FORMING TOOLING DATA PROVISIONAL. TO BE CONFIRMED BY MGI ENGINEERING.

BILL OF MATERIALS

Item	Description	Material	Qty	kg ea
1	Bellows, U-36, 1 × 1.5 mm, N=4, Ø600 ID	SA-240 321	1	9.6
2	Pipe nipple Ø600 × 9.53 × 150 lg	A106 Gr B / A53	2	20.8
3	Slip-on flange Ø813 × 46, 20 × Ø35	A105 (Cl. 150)	2	77.9

TOTAL APPROX. MASS 207 kg (EMPTY)

PRELIMINARY — FOR QUOTATION / ENGINEERING REVIEW

PT MISUBA GUNA INDONESIA

SINGLE EXPANSION JOINT — FLANGED, Ø600 BORE

EJMA design ref. EJ-2026-6BMPAW

DWG NO.	**MGI-EJ-2026-6BMPAW-01**	REV	**A**
SCALE	**1:6 / DETAIL 2:1**	SHEET	**1 OF 1**
UNITS	**mm**	DATE	**2026-10-13**

DRAWN: _____ CHECKED: _____ APPROVED: _____

Third angle projection. Generated from EJMA calculator result EJ-2026-6BMPAW and Fusion model. Do not scale.